

FEPrepare: A Web-Based Tool for Automating the Setup of Relative Binding Free Energy Calculations

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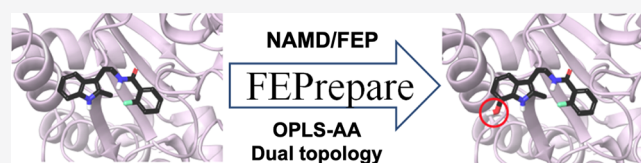


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ABSTRACT: Relative binding free energy calculations in drug design are becoming a useful tool in facilitating lead binding affinity optimization in a cost- and time-efficient manner. However, they have been limited by technical challenges such as the manual creation of large numbers of input files to set up, run, and analyze free energy simulations. In this Application Note, we describe FEPrepare, a novel web-based tool, which automates the setup procedure for relative binding FEP calculations for the dual-topology scheme of NAMD, one of the major MD engines, using OPLS-AA force field topology and parameter files. FEPrepare provides the user with all necessary files needed to run a FEP/MD simulation with NAMD. FEPrepare can be accessed and used at <https://feprepare.vi-seem.eu/>.



INTRODUCTION

Computer-aided drug design has become an integral part of drug discovery and is nowadays extensively used in the lead identification and lead optimization phases.^{1–3} Following the identification of active compounds for a target of interest, lead optimization includes improving candidate drug solubility, oral bioavailability, metabolic stability, and, importantly, its potency/binding affinity. One of the most important tasks in lead optimization is to predict among a series of lead candidates which ones will bind more strongly to the therapeutic target. The process of delivering optimized leads of higher affinity than the parent compound and providing coherent structure–activity relationships that can efficiently guide synthetic efforts can now be both time- and cost-efficient using reliable computational methods to calculate protein–ligand binding affinities.³ Relative binding free energy calculations can be performed using physics-based molecular simulations and rigorous statistical mechanics to calculate the differences in the free energy of binding between a parent candidate drug and its analogues.⁴ Over the past decade, improvements in force fields and sampling algorithms together with advancements in computing power, technology, and resources have enabled the fast and efficient implementation of rigorous theoretical methods to compute relative and absolute free energy of binding of a ligand to a protein target,⁴ such as free energy perturbation (FEP)⁵ and thermodynamic integration (TI),⁶ with significant successes in prospective lead optimization.^{7–17} An accurate force field is a necessary condition obtaining reliable free energy predictions; years of academic and industrial research have delivered force fields that can generate relative binding free energies at a level of accuracy useful in drug discovery.^{18–21} Moreover, unprece-

ded CPU and GPU speedups have emerged since the first FEP calculations^{22,23} were performed. A decade ago, an MD engine could obtain ~1 ns/day on a single modern CPU for a solvated ~25k atom protein system, whereas now ~1000 ns/day can be reached, e.g., using AMBER16 with GPU acceleration for the same protein system.^{4,24}

However, the technical challenges to set up, run, and analyze FEP/TI simulations have hampered the wider adoption and throughput of using these methods in drug discovery.¹⁰ Automating FEP calculations is a step forward to delivering high throughput calculations for accurate predictions of relative binding affinities before a compound is synthesized and consequently save enormous time and cost.

Simulation packages still offer limited support for setting up relative free energy simulations, although some efforts geared toward automating the setup of free energy calculations have been made. Commercially available automated FEP/TI pipelines include the FEP+ module of the Schrodinger Suite,²⁵ the Molecular Operating Environment (MOE) offered by the Chemical Computing Group (CCG)¹⁸ providing a pipeline for performing relative binding free energy calculations using the AMBER TI platform,²⁶ BIOVIA Discovery Studio and Pipeline Pilot²⁷ offering a workflow for setting up and implementing relative FEP calculations on GPUs using CHARMM,²⁸ and FEP in Flare²⁹ of the Cresset software that

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uses open-source tools such as AMBER tools,³⁰ OpenMM,³¹ LOMAP,³² Sire,³³ and BioSimSpace.³⁴ Noncommercial workflows include the Free Energy Workflow tool³⁵ available for AMBER,³⁰ QligFEP³⁶ for the Q open-source MD package,³⁷ FESetup²⁵ for AMBER and GROMACS,³⁸ BioSimSpace³⁴ that automates the creation of the required input files for the implementation of relative binding free energy simulations using SOMD³³ or GROMACS,³⁸ BFEE2³⁹ for absolute free energy calculations generating configurational files in NAMD⁴⁰ and GROMACS³⁸ formats, and two web servers, the “Alchemical Assistant” of the LigParGen server⁴¹ and the CHARMM-GUI FEP calculator.⁴² The CHARMM-GUI FEP calculator provides input and postprocessing scripts for NAMD⁴⁰ and GENESIS⁴³ and supports at the moment only the CHARMM force field CGenFF for FEP/MD calculations with NAMD.²¹ The “Alchemical Assistant” feature of the LigParGen server can provide input files for FEP calculations using NAMD.⁴¹ The LigParGen server parameterizes ligands using the OPLS-AA force field for small organic molecules with 1.14*CM1A or 1.14*CM1A-LBCC partial atomic charges, thereafter referred to as OPLS-AA.^{41,44,45} The files generated by the Alchemical Assistant follow a dual-topology approach, where in the topology file both reference and end state atoms exist simultaneously but do not interact with each other, and at the two end states of the FEP calculation one of the two ligands is fully interacting with the environment, while the other is noninteracting. In order to perform a FEP calculation with the Alchemical Assistant/LigParGen output using the OPLS-AA force field, the parameters of all atoms of the one end state need to be linearly scaled in order to reach the other end state atoms during the simulation. This could lead to slow convergence of the FEP/MD simulations due to the large number of perturbations that need to take place during the course of the simulation.

In this Application Note, we present a web-based tool, FEPPrepare, which automates the setup procedure for performing FEP/MD simulations using NAMD and the OPLS-AA force field. FEPPrepare uses a hybrid single/dual-topology protocol, allowing the assignment of a single- and a dual-topology region at the same time. In this approach, the nonperturbed atoms of the simulation are simulated using the single-topology approach, while the perturbed atoms are represented using the dual-topology protocol. Therefore, the linear scaling of the parameters of the atoms from one end state to the other is performed only for the perturbed atoms. This reduces the number of perturbations needed to take place during a FEP/MD calculation and leads to faster convergence of the FEP calculations using the OPLS-AA force field. FEPPrepare does not require additional libraries or software requirements and can be easily used as a web server for free and without a login requirement.

METHODS

FEP Theory. In FEP, the difference in the free energy of binding between two ligands A and B against a protein target is computed. Using MD simulations as the sampling technique one can compute the free energy change of transforming A to B through Zwanzig’s formalism⁵

$$\Delta G(A \rightarrow B) = -k_B T \ln \langle \exp(-\beta[U_B(q) - U_A(q)]) \rangle_A \quad (1)$$

where the angular brackets indicate that the quantity inside is averaged over all configurations of A and weighted by

Boltzmann probabilities. It is also possible to perform the reversible process, $\Delta G(B \rightarrow A)$, as well. In the limit of infinite sampling, the free energy changes between the two processes must be equal. However, in practice, this is not usually the case due to the asymmetry in the convergence rate of the free energy estimate to the true free energy change. This asymmetry can be understood in terms of the phase space overlap between the configurations of A and B that contribute the most to the total free energy estimate. For eq 1 to work properly, there should be an adequate overlap between the low energy configurations of A and B.

One way to achieve this overlap is to multistage the transformation of A to B. This is feasible by introducing a series of N alchemical intermediate states by defining a coupling parameter λ , such that the potential energy function $U(\lambda)$ interpolates between the two protein–ligand complexes of interest, A ($\lambda = 0.0$) and B ($\lambda = 1.0$). The result is that the exponential averaging is only implemented between states that have a high degree of phase space overlap and the overall free energy is estimated through the following equation:

$$\Delta G(A \rightarrow B) = \sum_{N=0}^{N=n-1} \Delta G(\lambda_N \rightarrow \lambda_{N+1}) \quad (2)$$

A further development of the multistage approach is the use of the thermodynamic cycle (Figure 1). The Gibbs free energy is a state function and, thus, the sum of all changes along a closed thermodynamic cycle is equal to zero. The use of the thermodynamic cycle allows to predict the difference in free energy of binding between ligands A (ΔG_A) and B (ΔG_B), which is computationally demanding, by computing the free

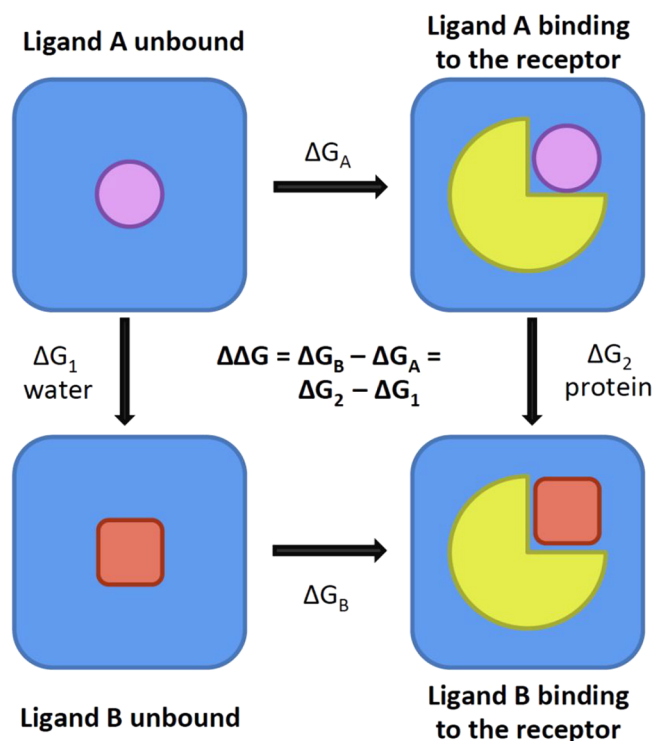


Figure 1. An example of the thermodynamic cycle used in relative free energy of binding calculations. The relative binding free energy of ligand B with respect to ligand A ($\Delta G_B - \Delta G_A$) is equal to the transformation of one ligand to another ($\Delta G_2 - \Delta G_1$) in water and in the protein binding pocket.^{4,46}

energy of transforming ligand A to ligand B in both solvent (ΔG_1) and bound (ΔG_2) phases. Therefore, the relative binding free energy difference between the two ligands ($\Delta\Delta G$) is computed via the equation³

$$\Delta\Delta G = \Delta G_B - \Delta G_A = \Delta G_2 - \Delta G_1 \quad (3)$$

For more information on the FEP theory, please refer to the [Supporting Information](#).

Single/Dual-Topology Approaches. A critical step in relative free energy of binding simulations is to choose the appropriate coupling technique, a mathematical definition that allows the smooth modification of the potential energy function of ligand A, U_A , to the potential energy function of ligand B, U_B . There are two major approaches, the single-topology and the dual-topology approach.

In the dual-topology approach, both reference and target state atoms exist simultaneously without interacting with each other, and while reference state atoms disappear, target state atoms appear.⁴⁷ At the final end state ($\lambda = 1.0$), the reference molecule is decoupled from its environment, while the other molecule (which we will call “perturbed” molecule hereafter) is fully interacting with the environment. The co-existence of the two molecules can be achieved by merging the reference ligand with the perturbed molecule into one file. In addition, spatial restraints need to be applied to the reference molecule to compensate for the fact that at the final end state the reference molecule is noninteracting and can drift apart from the receptor.⁴⁸ In the single-topology approach, the single-topology region contains only one molecular entity that consists of atoms from both ligands. The perturbed atoms from the one end state are transformed to the corresponding atoms of the other end state using the alchemical coupling parameter, λ . If the number of atoms is different between the two end states, a number of noninteracting dummy atoms is introduced to occupy the space of the appearing/disappearing physical atoms in order to balance the number of atoms at both end states.⁴⁰ In [Figure S1](#) both techniques are schematically presented. Many FEP/MD codes such as AMBER and GROMACS implement hybrid approaches, allowing the assignment of a single- and a dual-topology region at the same time. NAMD allows both the hybrid and the dual-topology scheme for FEP/MD calculations.⁴⁰

Data Structures. The partial charges and the force field parameters of small molecules can be generated through a plethora of software and web servers. In this paper, the LigParGen web server, which provides the OPLS-AA parameters for small organic molecules using two CM1A charge models, 1.14*CM1A and 1.14*CM1A-LBCC, is used.⁴¹ For the calculations, the acquisition of coordinate files (PDB), topology files (RTF), and parameter (PRM) files is required.

■ RESULTS

FEPPrepare NAMD/FEP Workflow. Alignment of ligands to a reference pose prior to performing relative binding free energy simulations is crucial for obtaining accurate predictions. Therefore, before conducting FEP calculations, all ligands should be aligned to the reference ligand (ideally a crystal structure). If multiple crystal structures are available, the crystal structure with the ligand most similar to the desired perturbation should be used. Ligand alignment should be conducted by the user before using FEPPrepare. For the alignment, the user can use maximum common substructure (MCS) superposition, and then, it is recommended to perform

postalignment minimization or a short MD equilibration, as after alignment the new ligand functional groups may present bad contacts with the protein structure. A combination of MCS alignment for the core and sampling of the modified substituents may offer the best balance for initial pose placement.^{4,46} For best practices in running FEP calculations, we refer the user to the review in [ref 4](#).

Parameterization of Ligands Using the LigParGen Server.

After the aligned ligand poses are obtained, the user needs to choose the desired charge model, define the molecule charge, and upload the molecules to LigParGen. Afterward, the required PDB, RTF, and PRM files can be downloaded; these output files are then uploaded to FEPPrepare as input. The automated FEPPrepare pipeline steps are explained below.

Atom Renaming. To avoid any inconsistencies in the atom naming that might occur and could cause NAMD to terminate abnormally, a Python script, which renames all atoms of the reference and perturbed ligands according to their coordinates, is implemented. The script creates six new files, three for each ligand (PDB, RTF, PRM), all renamed properly.

Partial Charge Redistribution. In the NAMD dual-topology scheme ([Figure S1](#)), both the reference and perturbed ligands exist in the same topology file (hybrid RTF file). In order to avoid perturbing all atoms of the reference ligand to the target ligand, an algorithm needs to be applied in order to reduce the required perturbations of atoms during the transformation and to speed up the convergence of the algorithm. Using our scheme, the common moiety between the reference and perturbed ligands retains the same parameters during the simulation, and only the perturbed area is perturbed. This is based on the assumption that in classical MD simulations the fixed point charge model is implemented, which cannot lead to changes in ligand partial charges. Finally, it should be noted that FEPPrepare works only when maintaining the same charge within the perturbation.

The workflow used for the creation of the hybrid RTF file is illustrated in [Figure S2](#). If the above workflow is utilized, then the perturbed area in both ligands should have the same charge in order for the total charge of the two molecules to be the same. For this to be achieved, a partial charge redistribution scheme needs to be employed, which is depicted in [Figure S3](#). Let Sum A be the sum of partial charges of the atoms that are being modified in the reference ligand and Sum B be the sum of partial charges of the atoms that are being modified in the perturbed ligand. For these sums to be equal, their difference called “extra charge X” needs to be distributed to the perturbed atoms and to the neighboring-to-the-perturbation atoms in the perturbed ligand. To distribute the extra charge X, we need to divide it with the number of atoms (N) that take part in the perturbation. The quotient of this deviation, named DIV, is the charge that we want to dispense among the perturbed atoms. In addition, DIV acts as a cutoff threshold for the convergence of the FEP algorithm, as it needs to be negligible (less than 0.02). The $DIV = 0.02$ has been chosen to be 20% of the smallest charge in the topology NAMD file, which is the hydrogen atom of a methyl group with a charge of 0.1. If DIV is larger than 0.02, then the algorithm is not converged, and we need to include more atoms in our calculation. The inclusion of atoms is done recursively by choosing atoms from the reference and the perturbed ligands that have a smaller difference in their partial charge. Once the new atoms are selected, we repeat the calculations until DIV is smaller than 0.02. Now the extra charge has been distributed equally, and

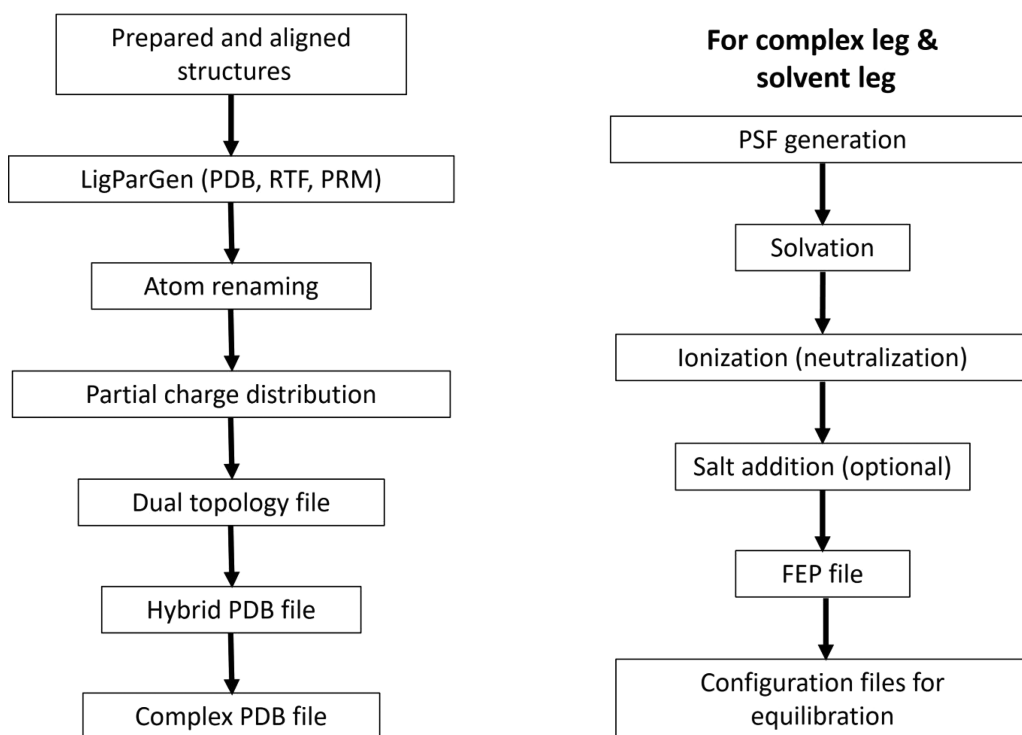


Figure 2. FEPrepare workflow.

thus, the atoms of the perturbed ligand that have taken part in the calculation have new partial charges. The implication of this charge re-distribution is that the charge differences between the two ligands now affect only the perturbed area atoms and not the whole ligand. These charge changes were tested by using DIV in a set of 20 perturbations in order to examine the spread of the charge and the number of atoms required to change in order to achieve this threshold (Table S1, Figure S4).

The description of the structure of the hybrid RTF file as well as the creation of the hybrid RTF and PDB files and the ligand-protein complex files can be found in the Supporting Information.

Complex and Solvent Leg Systems PSF Generation, Solvation, and Ionization Using VMD. The preparation of the solvent and ligand-protein complex systems can be performed through the VMD GUI;⁴⁹ FEPrepare uses an in-house Tcl script that calls VMD packages *psfgen*, *solvate*, and *autoionize* to create the PSF, solvate the system in a water box of 15 Å distance from the atom with the greatest coordinate in each direction, and insert counterions to neutralize the system, respectively. The user is also given an option to add 150 mM NaCl to the system as the most common option of salt physiological concentration. In addition, the script measures the minimum, maximum, and center of the box coordinates, which are needed for setting up the NVT equilibration run.

FEP Files Update and Correction of Two-Character-Named Chemical Elements Error in PDB Files. During FEP/MD calculations, NAMD uses FEP files (.fep) to identify which atoms disappear, which atoms are not modified, and which atoms appear during the simulation. The FEP file used here is a copy of the *ionized.pdb* file with a modification in the column dedicated for B-factor values. Values in this column are used to specify the atoms that will be perturbed during the FEP/MD calculation based on the partial charge redistribution

scheme. This is done by setting the values of the disappearing atoms to -1.00 , the values of the unmodified atoms to 0.00 , and the values of the appearing atoms to 1.00 .

The algorithm also corrects the following error that VMD produces. For atoms named with two letters (such as Cl or Br) VMD fails to read their coordinates during PSF generation, and instead prints 0.000 in x , y , and z coordinates in the *ionized.pdb* file. The FEPrepare algorithm corrects this error by reading the correct coordinates from the initial PDB files and updating the *ionized.pdb* file. This algorithm is explained in Figure S5.

WEB SERVER IMPLEMENTATION

Development. For the above-mentioned methodology, a new algorithm was programmed in Python and no previously developed algorithms were reimplemented. The workflow of FEPrepare can be seen in Figure 2. PHP runs in the back-end in the current web server implementation, and the front-end is written in HTML, CSS, and JavaScript. A user manual, a video example, and a test case are included on the website.

User Provided Input. FEPrepare can handle input from LigParGen. The coordinate PDB files of the two ligands and the protein, the LigParGen topology, and the parameter files of both ligands (reference and perturbed ligand) need to be uploaded. The ligand coordinates need to be aligned before uploading with respect to both each other and the protein.

Output. FEPrepare provides all files necessary to run NAMD/FEP as well as all files generated during the FEPrepare procedure for full transparency and in case the user wants to modify any inputs along the process. These files are explained in the README file that accompanies the output. A sample README file is provided in the Supporting Information.

Case Studies. After the user uploads the files needed for FEPrepare, the algorithm downloads the files, reads them, and renames atoms to avoid inconsistencies. Then, the algorithm

continues with the partial charge distribution and the creation of the dual-topology file. Here, we have implemented two case studies with FEPPrepare using outputs to perform FEP/MD calculations using NAMD 2.14.

In the first case, we use actin-related protein 2/3 (Arp2/3) complex, a seven-subunit assembly that generates branched actin networks,⁵⁰ which deregulation can promote cancer progression and, thus, inhibition of this protein with small molecules can control this progression.⁵¹ The second case, cyclin-dependent kinase 8 (CDK8), is a transcriptional kinase that regulates the WNT/ β -catenin signaling pathway leading to colon tumorigenesis⁵² and can be a promising target for cancer therapy.⁵³ More information regarding these proteins can be found in the [Supporting Information](#).

In the first case study for the Arp2/3 protein, we examined the transformation of a known small inhibitor of Arp2/3 (CK666) against a perturbed ligand (AI003). In [Figures S6 and S7](#), it is shown that all atoms in the reference and target ligands are the same, apart from atoms **H86** (in CK666) and **O0N** and **H14** (in AI003). This is the area that will be perturbed in the ligand; a hydrogen atom will be perturbed into a hydroxyl group. This modification introduces an extra charge in the area of the perturbation. The algorithm considers the extra charge and merges the atoms of the reference ligand (CK666) with the atoms that take part in the perturbation of the perturbed ligand (AI003), in our case **O0N** and **H14**. The *hybrid.rtf* and *hybrid.pdb* files are then created, and the PRM file of AI003 is updated. The next step is to combine the protein PDB file with the *hybrid.pdb* file in order to create the *complex.pdb*. The code generates correctly the files that VMD needs to run its calculations and creates the PSF files.

The output files are then provided to NAMD, and the calculation is run (see [Supporting Information](#) files). These output files are compatible with NAMD, since the simulation finishes correctly and provides the same results as when the simulation was run with manually created inputs (see [Supporting Information](#) files).

In the second case study for the CDK8 protein, we examined the perturbation between inhibitors 34 and 36 ([Figures S8 and S9](#)) using FEPPrepare.⁵² The server generates correctly the required input files for the FEP/MD calculations.

We also performed FEP/MD calculations for the CK666–AI003 perturbation of the Arp2/3 complex using the Schrödinger FEP+ method⁷ with Schrödinger suite version 2018-3. The ligands were parametrized using the OPLS3e force field.¹⁸ Regarding the ligand 34 > ligand 36 transformation of the CDK8 case study, relative FEP calculations were implemented by Schindler et al. using also the same FEP+ protocol.⁵⁴ We compared the results from these calculations with relative binding free energies obtained from NAMD 2.14 and from experimental data ([Table S2](#)). Finally, we plotted the convergence plots for the calculated binding free energies in solvent and complex legs of the CK666–AI003 perturbation obtained from FEP+ ([Figure S10](#)) and NAMD ([Figure S11](#)) as a function of time.

The FEPPrepare server deletes the user input and the resulting output files 2 days after the upload of the PDB, RTF, and PRM files. In addition, a new user will not be able to view the files uploaded by other users.

CONCLUSIONS

The strength of the binding affinity of a small molecule binding to a target of interest is a necessary condition for drug potency

alongside other critical factors such as solubility, residence time, modifying a downstream pathway, optimizing for delivery, and others. MD simulations, including enhanced sampling techniques, can be useful in the hit-to-lead and lead optimization stages in the drug design process.^{55–57} MD-based methods can be used to study the stability of a protein–ligand complex and to model conformational changes in the biological target that may occur upon ligand binding.^{58–60} In addition, MD simulations are frequently employed as a sampling technique for the calculation of free energy differences between ligands of congeneric series using FEP^{61–63} and with simulation packages such as AMBER, GROMACS, and NAMD, although other sampling techniques such as Monte Carlo are also applied.⁶⁴

Although NAMD is widely used, its FEP setup procedure with the dual-topology scheme has not yet been streamlined with a variety of force fields. This motivated us to create a web-based server that automates the FEP/MD dual-topology procedure using the OPLS-AA force field to provide fast and automated input NAMD FEP/MD files ready for production runs. Before the implementation of FEPPrepare, the manual creation and manipulation of NAMD/FEP input files was a rather tedious and time-consuming process amounting to approximately a full day of work for a single-reference-target ligand pair for an experienced user. FEPPrepare automates the setup procedure and creates all necessary inputs in seconds. In addition, it decreases the computational cost needed to perform dual-topology FEP/MD calculations.

The dual-topology protocol was implemented in NAMD by merging the reference ligand and the perturbed ligand into one file and by linearly scaling the parameters of all atoms of the reference state to reach the target end state. This implementation leads to slower convergence of the FEP/MD simulations, thus requiring increased amounts of computational time. FEPPrepare solves this problem by introducing a hybrid single/dual-topology protocol that reduces the number of perturbations needed for FEP/MD calculations and therefore leading to faster convergence of the simulations. Finally, FEPPrepare allows the user to perform FEP/MD calculations using the OPLS-AA force field for the parametrization of small organic molecules; parameters are provided by the LigParGen Web server.

Ultimately, there are a few further implementations, which can expand the use of FEPPrepare in the future such as the implementation as a stand-alone code. Furthermore, the server works only for one pair of ligands; setting up multiple ligand pairs could be envisaged by including maximum common substructure ligand perturbation maps with the use of tools such as LOMAP,³² which reduces the graph of all possible ligand pairs to a minimum set based on a definition of similarity used to weigh the graph edges to solve the shortest path tree problem. In such a map, cycle closures between ligands are also built in order to check for consistency of the relative FEP results. Finally, parametrization of small molecules with other force fields such as GAFF2²⁰ could also increase the applicability of FEPPrepare.

Overall, this work led to the creation of a novel web-based tool that automates the setup of all necessary input files for relative FEP calculations using the dual-topology approach of NAMD and the OPLS-AA force field. It is hoped that the automation of FEP/NAMD calculations through FEPPrepare will be useful in the adoption of relative binding free energy

calculations by the scientific community and that it will be widely applied in drug discovery projects.

■ ASSOCIATED CONTENT

SI Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jcim.1c00215>.

Input and output files for the manually created Arp2/3 use case (ZIP)

Input and output files for the Arp2/3 use case (ZIP)

Input and output files for the CDK8 use case (ZIP)

Additional information and supporting figures and tables describing the FEP theory, FEPPrepare workflow, methodology and README file and results of use cases (PDF)

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Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

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Notes

The authors declare no competing financial interest.

Data and Software Availability. All calculations were performed with freely available tools and codes. All novel methods were comprehensively described in the Methods section, in relevant workflows and the Supporting Information. The Web server is freely accessible at <https://fepprepare.vi-seem.eu/> and the code in <https://github.com/zoecournia/FEPPrepare>. All relevant data, including all molecular structures and input and output files, are available in a machine-readable format in the Supporting Information. FEPPrepare can be used freely under the condition that VMD is installed in the user's personal computer.

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